EXPERIMENT 3:

```
clc;

 clear;

 close all;

% Parameters dt = 0.001; t = 0:dt:5;

% Energy Signal xE = exp(-t);

% Power Signal xP = sin(2pi5*t);

% Numerical Energy Calculation E_num = trapz(t, abs(xE).^2);

% Numerical Average Power Calculation T0 = 1/5; idx = t <= 2;

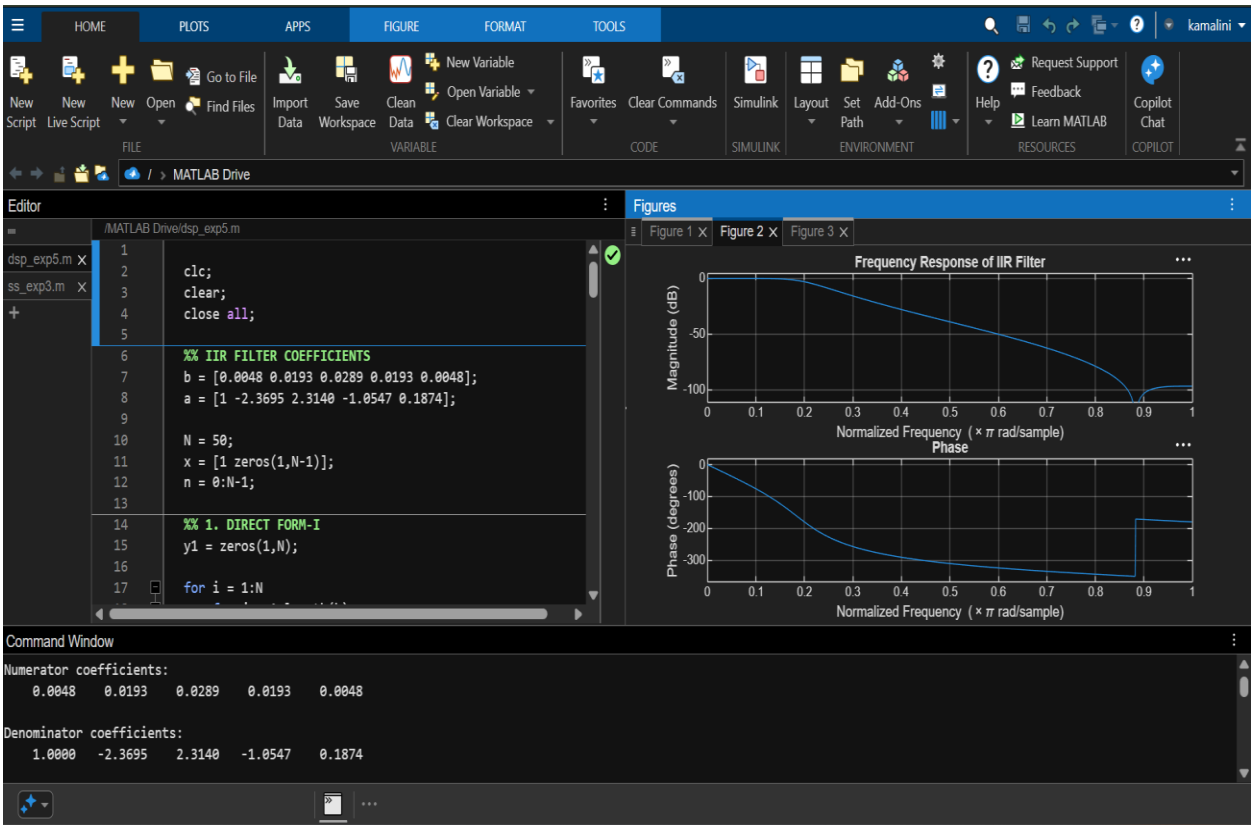
P_num = (1 / (t(find(idx, 1, 'last')) - t(1))) * ... trapz(t(idx), abs(xP(idx)).^2);

% Plotting figure;

subplot(2,1,1); plot(t, xE); grid on; title('Energy Signal e^{-t}u(t)'); xlabel('Time (s)');
ylabel('Amplitude');

subplot(2,1,2); plot(t, xP); grid on; title('Power Signal sin(2\pi5t)'); xlabel('Time (s)');
ylabel('Amplitude');

% Display Results fprintf('Numerical energy = %g\n', E_num); fprintf('Numerical average power
= %g\n', P_num);
```


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FIGURE
FORMAT
TOOLS
kamalini
New Script
New Live Script
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Open
Go to File
Find Files
FILE
Import Data
Save Workspace
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New Variable
Open Variable
Clear Workspace
VARIABLE
Favorites
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CODE
Simulink
SIMULINK
Layout
Set Path
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ENVIRONMENT
Help
Request Support
Feedback
Learn MATLAB
RESOURCES
Copilot Chat
COPILOT
/ › MATLAB Drive
Editor
/MATLAB Drive/dsp_exp5.m
dsp_exp5.m
ss_exp3.m
clc;
clear;
close all;
%% IIR FILTER COEFFICIENTS
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];
N = 50;
x = [1 zeros(1,N-1)];
n = 0:N-1;
%% 1. DIRECT FORM-I
y1 = zeros(1,N);
for i = 1:N
Figures
Figure 1
Figure 2
Figure 3
Frequency Response of IIR Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
Numerator coefficients:
0.0048 0.0193 0.0289 0.0193 0.0048
Denominator coefficients:
1.0000 -2.3695 2.3140 -1.0547 0.1874